

Assignment 4

Solve the following problems and upload the file of solutions in the D2L. You may type the solutions in Microsoft Word and upload the file, or you may scan your solutions by scanner and upload the file in the D2L (**due date: Wednesday, November 22, 11:59 am, no extension!**).

1. (20 points)

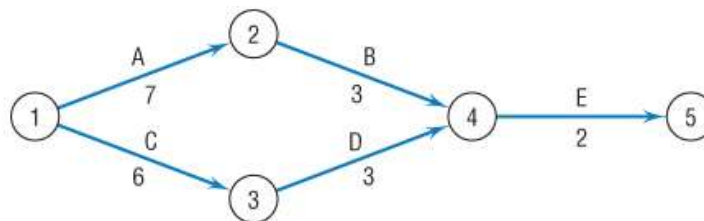
- What is an activity slope and what does it indicate?
- What are two methods for addressing the constrained resources allocation problem?

2. (20 points)

- What is the purpose of “earned value”? How would the progress of the project as a whole be calculated?
- What are the primary difficulties experienced in the design of project reports? Mention four of them.

3. (20 points)

Consider the following network for conducting a 2-week (10 working days) computer training class:



Activity	Crash Time (days)	Crashed Cost (total)	Normal Time (days)	Normal Cost
A: Obtain instructor	4	\$800	7	\$500
B: Locate a room	2	350	3	200
C: Check costs	4	900	6	500
D: Room open?	1	500	3	200
E: Schedule class	1	550	2	300
		\$3100		

- Find the critical path.
- Find the crash cost per day.
- Which activities should be crashed to meet a project deadline of 10 days with a minimum cost? Assume partial crashing.
- Find the new cost.

4. (20 points)

Consider the following activity information and the constraint that the project must be completed in 16 weeks.

Activity	Prec. Evt.	Suc. Evt.	TE (weeks)	Prec. Activ.
a	1	2	3	None
b	1	3	6	None
c	1	4	8	None
d	2	5	7	a
e	3	5	5	b
f	4	5	10	c
g	4	6	4	c
h	5	7	5	d,e,f
i	6	7	6	g

In addition, activities **c**, **f**, **h**, and **i** may be crashed as follows. Assume partial crashing.

Activity	Crash Time (weeks)	Additional Cost per Week
c	7	\$40
f	6	20
h	2	10
i	3	30

Find the best schedule and its cost.

5. (20 points)

a. A software development project at day 70 exhibits an actual cost of \$78,000 and a scheduled cost of \$84,000. The software manager estimates a value completed of \$81,000. What are the cost and schedule variances and CSI? Estimate the time variance.

b. A consulting project has an actual cost in month 10 of \$23,000, a scheduled cost of \$17,000, and a value completed of \$20,000. Find the schedule and cost variances and the three indexes.